AP BIOLOGY TOPIC 7.5

Hardy-Weinberg Equilibrium & Null Model of Evolution

UNIT 7: NATURAL SELECTION (13–20% EXAM WEIGHT)

EVO-1.K | EVO-1.L | P + Q = 1 | P² + 2PQ + Q² = 1

Credits: mitali120501@gmail.com

§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 7.5 (HARDY-WEINBERG EQUILIBRIUM & NULL MODEL OF EVOLUTION)

CED Core Vocabulary

Theoretical Null Model — Mathematical model describing a hypothetical non-evolving population in which allele and genotype frequencies remain constant from generation to generation.

Single-letter variables = ALLELE frequencies — (gamete pool, total = 1.0 or 100%): p = frequency of the dominant allele (A)

Golden Rule — When given phenotype counts, ALWAYS start with the homozygous recessive phenotype (q^2) because it is the only phenotype whose genotype is known for certain!

Step 5 (Number of Heterozygotes) — 0.48 × 500 = 240 heterozygous (Aa) plants !

Purpose of the Model — Acts as a baseline "control group" (null hypothesis H_0) for population genetics: Scientists compare observed genotype frequencies in nature against expected...

Dominant Phenotype Frequency — $p^2 + 2pq$ (or $1 - q^2$). Never set dominant phenotype equal to p^2 alone!

Step 4 (Find Genotypes) — $p^2 = (0.60)^2 = 0.36$ (AA) | $2pq = 2(0.60)(0.40) = 0.48$ (Aa carriers)

§2. WHAT IS HARDY–WEINBERG EQUILIBRIUM (HWE)?

Null Hypothesis Model

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

What Is Hardy–Weinberg Equilibrium (HWE)? ➔ **Theoretical Null Model** ➔ **Purpose of the Model**

- **Theoretical Null Model:** Mathematical model describing a hypothetical **non-evolving population** in which allele and genotype frequencies remain constant from generation to generation.
- **Purpose of the Model:** Acts as a baseline **"control group" (null hypothesis** H_0 **)** for population genetics:
  - Scientists compare **observed genotype frequencies** in nature against **expected Hardy-Weinberg frequencies** (often using a Chi-Square χ^2 test).
  - Real populations rarely meet all conditions; significant deviations indicate that one or more evolutionary forces are actively acting on the population!

[CORE AXIOM / REMEMBER] **Hardy–Weinberg = "Control Group" for Evolution.** If **observed ≠ expected** frequencies → **evolution is occurring**!

§3. THE 5 CONDITIONS REQUIRED FOR HARDY–WEINBERG EQUILIBRIUM

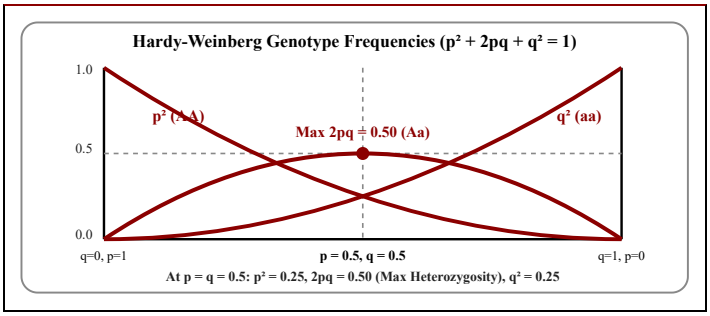
Must Memorize All 5

HWE CONDITION	EVOLUTIONARY FORCE PREVENTED	EFFECT IF VIOLATED
1. Extremely Large Population Size	Prevents **Genetic Drift**	Small **N** causes random chance fluctuations in **p** and **q**.
2. No Migration (Isolation)	Prevents **Gene Flow**	Immigration/emigration adds or removes alleles from the gene pool.
3. No Net Mutations	Prevents **Mutational Shift**	New alleles arise or convert one allele into another (**A → a**).
4. Random Mating	Prevents **Assortative Mating / Inbreeding**	Alters **genotype frequencies first** (e.g., inbreeding increases **p^2** & **q^2**, decreases **2pq**).
5. No Natural Selection	Prevents **Differential Survival / Fitness**	All genotypes have equal fitness (**w = 1**); selection shifts both genotype & allele frequencies.

[COMMON AP EXAM FRQ TRAP] **Non-random mating (e.g., self-fertilization or inbreeding) changes GENOTYPE frequencies (** **p^2, 2pq, q^2** **) first, NOT allele frequencies (** **p, q** **) directly!** Sexual selection (differential mating success), however, does alter allele frequencies.

§5. VISUALIZING GENOTYPE FREQUENCIES ACROSS ALLELE VALUES

HWE Parabola Graph



§6. STEP-BY-STEP HWE PROBLEM-SOLVING GOLDEN RULE

AP Math Workflow

1. Find q^2 (aa phenotype) → **2. $q = \sqrt{(q^2)}$** → **3. $p = 1 - q$** → **4. Compute p^2 & 2pq**

- **Golden Rule:** When given phenotype counts, **ALWAYS start with the homozygous recessive phenotype (** **q^2** **)** because it is the only phenotype whose genotype is known for certain!
- If given dominant phenotype % (e.g., 84% dominant), subtract from 100% first to get recessive phenotype (**$q^2 = 1 - 0.84 = 0.16$**).

[EXAMPLE & BIOLOGICAL APPLICATION] **Worked AP Calculation**

Problem:

In a population of 500 wildflowers in HWE, 80 plants have white flowers (recessive, ***aa*** *) and 420 have red flowers (dominant,* ***AA*** *or* ***Aa*** *). Find all allele & genotype frequencies and the number of heterozygous plants:*

- **Step 1 (Find q^2):** **$q^2 = 80 / 500 = 0.16$** (homozygous recessive genotype frequency)
- **Step 2 (Find q):** **$q = \sqrt{0.16} = 0.40$** (recessive allele frequency)
- **Step 3 (Find p):** **$p = 1 - 0.40 = 0.60$** (dominant allele frequency)
- **Step 4 (Find Genotypes):** **$p^2 = (0.60)^2 = 0.36$** (**AA**) | **$2pq = 2(0.60)(0.40) = 0.48$** (**Aa** carriers)
- **Step 5 (Number of Heterozygotes):** **0.48 × 500 = 240 heterozygous (Aa) plants** !

[AP EXAM FRQ SCORING STRATEGY] **Common AP Math Trap:** Read carefully whether the question asks for **"frequency of the recessive allele" (** **q** **)** or **"frequency of the homozygous recessive genotype/phenotype" (** **q^2** **)** — and whether it wants a decimal frequency (**0.48**), percentage (**48%**), or headcount (**240 individuals**)!

§4. THE TWO HARDY–WEINBERG EQUATIONS (AP FORMULA SHEET)

Alleles vs. Genotypes

1. Allele Frequencies: p + q = 1 | 2. Genotype Frequencies: p² + 2pq + q² = 1

- **Single-letter variables = ALLELE frequencies** (gamete pool, total = 1.0 or 100%):
  - **p** = frequency of the **dominant allele (A)**
  - **q** = frequency of the **recessive allele (a)**
- **Squared/Product terms = GENOTYPE (individual) frequencies** (diploid individuals, total = 1.0):
  - **p²** = frequency of **homozygous dominant (AA)** individuals
  - **2pq** = frequency of **heterozygous (Aa) carriers**
  - **q²** = frequency of **homozygous recessive (aa)** individuals (equals recessive phenotype frequency!)
- **Dominant Phenotype Frequency:** **p² + 2pq** (or **1 − q²**). Never set dominant phenotype equal to **p²** alone!

§7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 7.5 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 7.5: Hardy-Weinberg Equilibrium & Null Model of Evolution ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 1, 2 & 3 (Artificial Selection, Hardy-Weinberg Mathematical Modeling & BLAST Bioinformatics): Comparing allele frequency shifts across generations under directional selection vs. small-population genetic drift simulations, and constructing maximum-parsimony cladograms from BLAST percentage sequence identity matrices (Cytochrome c / β-globin). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Golden Rule: Hardy-Weinberg Equilibrium (HWE): Allele frequencies p + q = 1; Genotype frequencies p² (homozygous dominant) + 2pq (heterozygous carriers) + q² (homozygous recessive) = 1. Golden Calculation Sequence: ALWAYS begin with the recessive phenotype frequency (q²), take √(q²) to find q, compute p = 1 − q, and then calculate 2pq (heterozygote carrier frequency). Never take the square root of the dominant phenotype frequency (p² + 2pq)! **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Evolutionary & Biomedical Disruption Case Study: Incomplete antibiotic regimens or agricultural antibiotic overuse exert strong directional selection on bacterial populations (e.g., MRSA or MDR-Tuberculosis): susceptible cells die while rare, pre-existing resistant mutants (arising via random mutation or R-plasmid conjugation) survive and reproduce. In conservation biology, severe population bottlenecks (e.g., Northern Elephant Seals, African Cheetahs) drastically erode genetic diversity via genetic drift, increasing homozygosity for deleterious recessive alleles and vulnerability to novel pathogens.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 7.5 to prove why the claim holds.

Credits: mitali120501@gmail.com

Support this site
Scan to donate